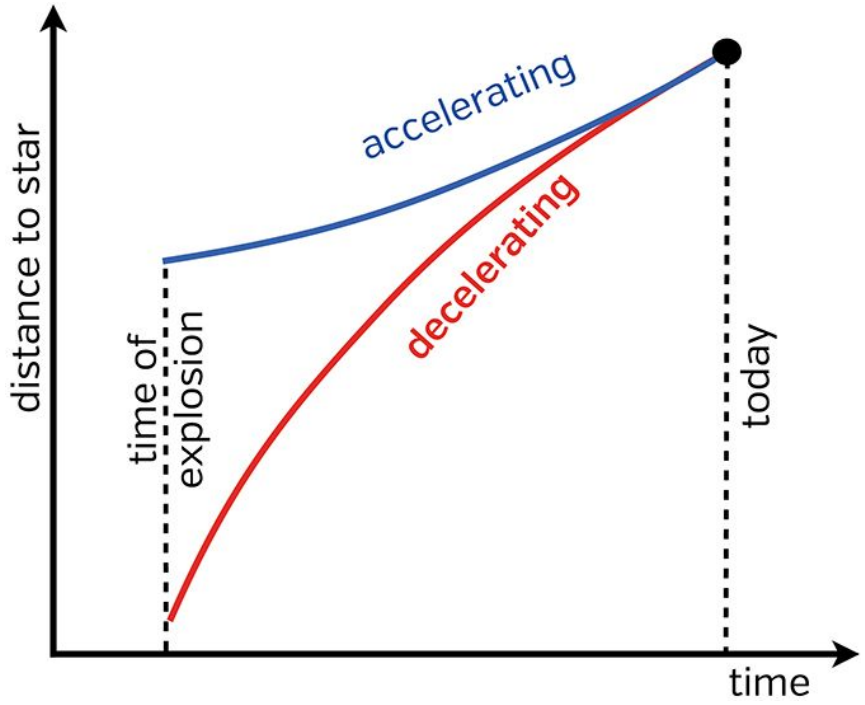


Fig. 24.4. The distance to the star at the time of explosion (the time that the light we see was emitted), under two assumptions: that the universe's expansion is decelerating (*red*) or accelerating (*blue*). The explosion was dimmer than expected, so farther away. The universe must be accelerating.

Now, our best understanding of gravity and the universe required, unequivocally, that all things in the universe (stars, galaxies, galaxy clusters, dark matter, etc.) must *pull* on each other gravitationally. And by that pull they must *slow* the universe's expansion. The universe's expansion must slow down over time, not speed up.

For this reason, I, personally, didn't believe the claimed acceleration, nor did many of my astronomer and physicist colleagues. We didn't believe until other observations, by completely different methods, confirmed it. Then we caved.

So what's going on? There are two possibilities: Something is wrong with Einstein's relativistic laws of gravity. Or something else is filling the universe, in addition to ordinary matter and dark matter. Something that *repels* gravitationally.